INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

JUNEJA STREET

Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

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Investment hypothesis and complete strategy

The strategy seeks short-term rebounds after unusually strong selling in volatile stocks. A fresh low alone is not a BUY: price must confirm an upward Supertrend reversal, recent oversold RSI and signal spacing. The machine-learning layer learns online from completed hypothetical long and short probes. Three model components use rolling outcomes, adverse-movement-weighted batches and an 8 by 8 market-context memory to adjust the Supertrend multiplier and ATR smoothing within bounds. The learning changes signal sensitivity and timing; explicit portfolio rules still set entry, position size and exit. The hypothesis fails when selling reflects lasting information, or costs erase any rebound.

Each probe opens at a candle open and matures at the close of its fifth candle. A move of one entry ATR gives the long probe a score of +1 and the matching short probe -1. Only matured outcomes can update later settings. Because probes overlap, their labels are dependent and are not account returns.

At equal confidence, one ATR of adverse movement halves a probe's batch weight. Context-memory influence rises with its supporting observations but is capped at 65%. A bounded update layer smooths the models' parameter proposals; it does not forecast a stock's return or decide how much capital to allocate. Position size follows separate risk limits.

Why these stocks

DELL is the paper-trading example. For risk calibration, we use a separate historical cross-sector sample. It is not the strategy portfolio or a list of declared holdings. The trading universe remains broader: each stock must pass its BUY, liquidity and risk checks before receiving capital.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for supported S&P 500 stocks. BUY opens one long position; repeated BUYs do not add size. SELL or an earlier protective stop closes it. Orders follow confirmation; historical replay fills at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$
$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the multiplier. The boundaries sit mA above and below x; the baseline uses l = 24 and m = 1.4. They trail or reset around price. A close across the previous opposing band flips trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

Three online model components use matured probes. The rolling-performance learner tracks a 30-outcome baseline window per direction: below 38% wins with negative mean score, it proposes wider bands and slower ATR smoothing; above 62% with a positive mean, tighter and faster settings. The micro-batch learner weighs each result by confidence divided by one plus adverse excursion in ATR units, then assesses normalized gain, profit factor and downside ratio. The context-memory learner maps trend/range state and relative volatility to an 8 by 8 grid. Hurst persistence, inverse entropy and ADX contribute to the regime score; smoothed cell proposals blend with Gaussian-distance-weighted neighbors. Its influence grows with sample count, capped at 0.65.

A meta-combiner blends rolling and context proposals, then incorporates available batch recommendations. A larger multiplier requires a larger reversal to flip trend; a longer ATR length reacts more slowly to volatility. The smoothed update changes signal timing, not account exposure. Deadbands, per-update caps, bounded parameter ranges and periodic baseline reversion limit drift. These are adaptive online-learning rules, not a neural network or evidence of predictive edge.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current multiplier or ATR length; theta tilde is the rounded, bounded target; alpha is the update fraction. For multiplier 1.40, target 1.50 and alpha 0.20, the update is 1.42. Deadbands ignore tiny changes; caps constrain each update and periodic reversion pulls settings toward baseline. The multiplier stays 0.5-5.0 and ATR length 5-100.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within 50 candles and at least 10 candles since the last signal. RSI compares smoothed gains and losses on a 0-100 scale. SELL uses a fresh-high reversal, downward flip and recent RSI above 70. Volume-surge filtering and live tick pressure are off; all decisions use confirmed candles.

Learning check. Across 780 DELL bars, 776 probes per side matured. The multiplier moved from 1.40 to 2.145 while ATR length remained 24; the model produced 6 BUY and 8 SELL signals. Prefix checks matched. This verifies the calculation sequence, not predictive performance. Data: Yahoo Finance.

2 Historical replay and backtest charts

We replayed the archived daily BUY rankings from 8 September 2025 to 30 October 2026 (290 sessions), starting with $100,000. Each saved ranking was applied at the next available session open, using the historical replay's position sizing and execution assumptions. The run held up to ten names; remaining positions were marked at the final close. This is a separate historical signal-ranking test, not a backtest of the declared 30-minute learning model.

Historical replay result	Value
Ending marked equity	$151,542.68
Total return / annualized growth	+51.54% / +43.92%
Maximum drawdown	-21.10%
QQQ benchmark return	+19.39%

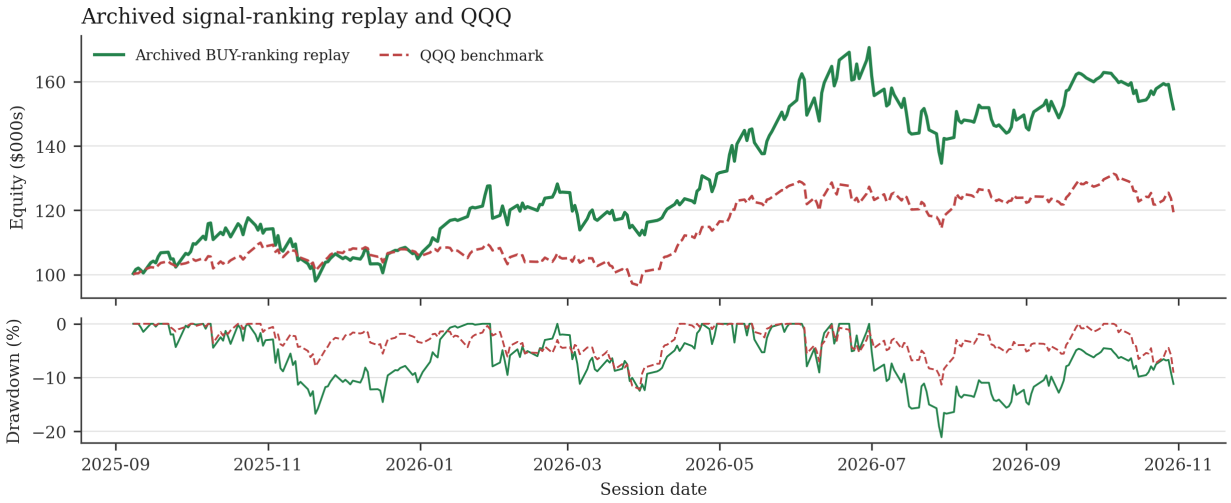


Figure 1. Archived ranking replay and QQQ benchmark: marked equity and peak-to-trough drawdown. Data: Alpaca SIP split-adjusted daily bars; our replay calculations.

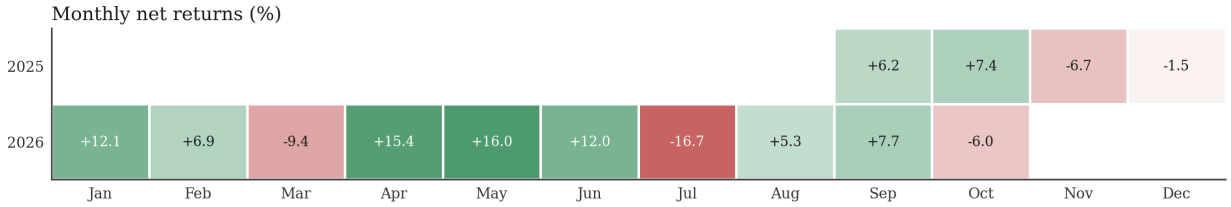


Figure 2. Monthly net returns (%); September 2025 is a partial month. Data: same replay.

Sensitivity: total return was +39.11% at 20 bps per side and +43.49% when ranking changes were delayed by one extra session. The replay reproduces the saved fills and daily marks; it does not establish future performance or the learning model's contribution. Rules, source hashes and reproduction steps: archived backtest documentation.

3 Trade expectancy and risk tests

Net Trade Expectancy Equation

$$e = pW - (1-p)D, \qquad p_{BE} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

Trade statistic	Historical result
Completed trades	622
Win rate	46.62%
Mean net profit / return per trade	+$81.73 / +0.8173%
Median trade return	-0.6268%
Average net winning / losing return	+8.05% / -5.50%
Profit factor	1.28
Break-even win rate at observed payoffs	40.59%
Worst trade	-40.03%
Longest losing streak	15 trades

These trade statistics come from the separate archived email-ranking replay, not the declared 30-minute model backtest. Within that earlier sample, the win rate exceeded the 40.59% break-even level implied by observed payoff sizes; profit factor was 1.28. The median trade still lost 0.6268%, showing that the mean gain was lifted by a small number of larger wins.

The 15-loss streak and -40.03% worst trade show why position limits and gap controls matter. These are 622 closed trades from the earlier email replay; ten open positions are excluded. Streaks follow recorded SELL order, including same-day exits. Calculations: GitHub trade evidence.

Monte Carlo: exposure, correlation and gaps

The separate historical risk sample is distinct from the five-position strategy portfolio. We fit its covariance to 59 aligned daily log returns, 2 July-25 September 2026, with 11.00% Ledoit-Wolf shrinkage. Each model runs 50,000 fifteen-session paths at 40%, 60% and 75% exposure, with constant quantities and zero cash interest. These scenarios compare risk levels; they are not strategy returns. Expected shortfall (ES) is the average account loss in the worst 5% of paths.

Risk scenario	ES 95% at 40% / 75%	Investment implication
Gaussian / Student-t(5)	3.50% / 6.56%; 3.56% / 6.68%	Zero-drift baseline / heavier tails.
Five-session block bootstrap	2.92% / 5.48%	Preserves sampled return blocks; unseen crashes remain possible.
Volatility x 1.5; correlation 0.90	9.36% / 17.55%	Correlated losses are the dominant portfolio stress.
Common 20% opening gap	10.81% / 20.27%	At 75%, daily-close liquidation lowers ES to 16.44%.

Data: Yahoo Finance daily closes; our calculations. Common dates only, no missing-price fills. Gaussian ES rises from 3.50% at 40% to 6.56% at 75%; the 60% comparison is 5.25%. At 75%, 7.18% of Gaussian paths reach the 6% drawdown trigger. These constant-holding scenarios test risk, not strategy returns. Gaussian expected price drift is zero; the short calibration leaves regime uncertainty. Seeds: 20261102-20261106.

4 Monte Carlo and stress charts

These figures expand the risk tests on page 4. The Monte Carlo dashboard summarizes 50,000 fifteen-session paths calibrated to 59 aligned daily returns. The lower chart applies simultaneous stock-price declines to the declared 75% exposure cap. Both are conditional risk scenarios, not strategy-return forecasts.

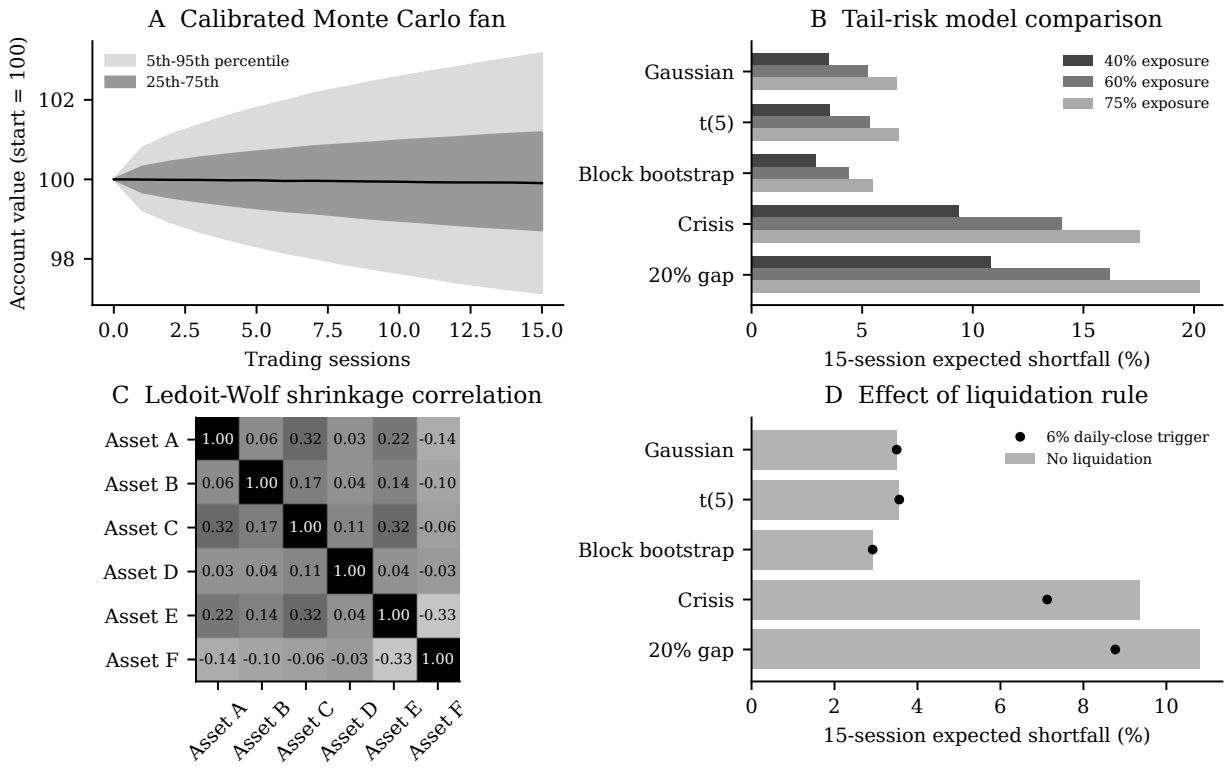


Figure 3. Panels A and D use 40% exposure; B spans 40%, 60% and 75%; C shows correlation for six anonymized assets. A is the 15-session value fan, B is ES95, and D tests the 6% close trigger. Data: Yahoo Finance; our calculations.

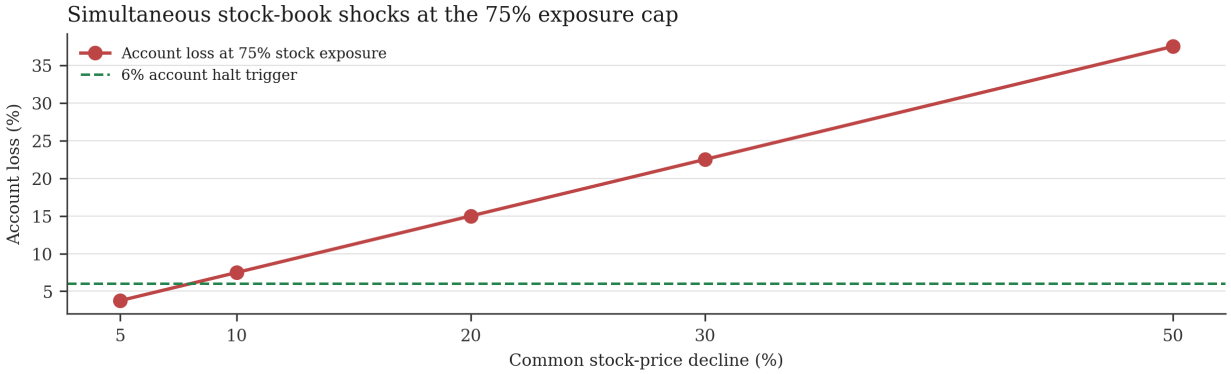


Figure 4. Deterministic common stock-book shocks at 75% gross exposure. The green dashed line marks the 6% account halt trigger. Stops, costs and recovery are excluded. Calculated from the declared exposure limit.

5 Portfolio construction and risk

False reversals can cluster during persistent downtrends; overnight gaps can cross stops, and technology names can fall together. Entry spacing, planned-loss budgets, sector/theme caps and the account halt address these risks. All limits below apply together and are declared design choices, not optimized thresholds.

Control	Execution rule and purpose
Exposure	Long-only, no borrowing; at most five names, 15% of equity per name and 75% gross stocks. Five names at the individual cap use the full portfolio allowance.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Execution integrity	Use one account; reject duplicate BUY entries. Log accepted/rejected signals, settings, quotes, order and fill times, fees, stop and exit reason. Freeze the declared rules; close by the competition cutoff.

Position Sizing Equation

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.15E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At the recorded DELL close of $563.17, a $15,000 cap permits 26 shares ($14,642.42). The $500 loss budget in a $100,000 account allows at most $19.23 stop distance at that size. A wider two-ATR stop reduces shares. The 15% name limit and five-position limit allow at most 75% gross exposure; remaining equity stays in cash. Position size can be lower when the stop-risk budget binds.

Gap and execution stress

Stop execution stress	Loss / planned stop loss	Account loss at 2% risk budget
Fills at intended stop	1.0 x	2%
Loss exceeds stop distance by 50%	1.5 x	3%
Loss doubles the intended distance	2.0 x	4%
Gap produces three times planned loss	3.0 x	6%

Calculation: aggregate planned loss multiplied by the realized loss/stop-distance ratio. This is a deterministic exposure test, not a forecast. Gaps can exceed all ratios shown; neither a stop nor the 6% liquidation trigger guarantees a maximum loss.

Stress grid: 96 combinations of horizon, exposure, volatility and correlation (100,000 paths each); daily Gaussian, t(5), gap and liquidation tests use 40,000 paths per model. At the 75% ceiling, common 10%/20% falls imply 7.5%/15% account losses before costs. A 6% halt is a trigger, not a loss guarantee.

Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold at most five names, 15% per name and 75% total stocks. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.